

UNIT 4: METOC Analysis Products

b. Surface Weather Observations

Introduction to Surface Weather Observations

Surface weather observations are coded data used to forecast weather conditions and issue weather warnings worldwide. Observations are recorded hourly. During periods of adverse weather conditions, they may be reported more frequently as needed. Observations are extremely important to military operations. They directly affect aviation safety, provide commanders vital information for mission impacts, and are the basis for determining and recording a locations climatology data. (scientific study of weather conditions over time).

The goal of this objective will be to show you have the capability to encode and decode a surface weather observation as well as perform the steps required to take a surface weather observation. To successfully demonstrate competency on this topic, you will be given information through lecture, a series of questions and tasks designed to display your ability to correctly identify and label various weather features discussed throughout this objective. These foundational analysis concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instruction for completing all requirements necessary to pass this objective.

Surface Weather Observation Reference Manuals

Meteorologists encode observations using the Meteorological Aerodrome Reports (METAR) code worldwide. The code has subtle differences between military branches and from one country to another based on their individual missions. However, the basic code remains uniform. Each branch has its own surface weather observation manual. These manuals are published by each branch and will be used throughout the rest of this course, as well as throughout your career as a weather forecaster. For any further guidance or information not covered in this text, navigate and reference the branch specific manuals.

- **Air Force - AFMAN 15-111**
- **Navy/Marines - NAVMETOCCOMINST 3141.2**

Exercise / Practice: Navigating the Manuals

Type of Observations

For meteorological observations the 'point of observation' is defined as the designated spot where the elements of an observation are viewed and/or sensed. The point of observation is within 5 statute miles of the airfield and affords as clear a view as possible of the runway complex.

METAR

METARs contain a complete report of wind, visibility, present weather, sky condition, temperature, dew point and altimeter setting collectively referred to as "the body of the observation." In addition, encoded and/or plain language information that elaborates on data in the body of the observation is appended at the end of the METAR in the remarks (RMK) section.

SPECI

Selected Special Weather Report (SPECI). A SPECI is an unscheduled observation completed and transmitted when any of the special criteria listed in the observation manuals are observed or sensed. SPECIs contain all data elements found in a METAR plus additional remarks that elaborate on data in the body of the observation. All SPECI reports will be prepared and transmitted as soon as possible after the relevant criteria are observed. The time ascribed to a SPECI reflects the time, to the nearest minute, that the SPECI criteria are first met or observed.

Automated Observations

An automated observation describes a type of observation that is solely based on the data gathered from the airfield sensor. The data is gathered at a set time and encoded into an observation based on only what the sensor displays. There is no input from the observer and no manipulation of the data it sends. This is usually the type of observation that is sent while the airfield is closed and there is not an observer present to augment the sensor data before submitting the observation.



Augmented Observations

Augmented observations are like automated observations in that they begin by pulling sensed data and formatting the observation, but it does not automatically submit the observation. Augmentation refers to an observer having the ability to go outside to the observation point and validate what the sensed data produced. Depending on what the observer sees, they may add, remove, or change the data given by the sensor before submitting the observation.



Manual Observations

Manual observations are exactly what the name implies. This type of observation requires the observer to go to the observation point and manually gather the required data, encode the data, and submit the observation without any input from the airfield sensor. Most elements of an

observation can be gathered just by an observer using their senses to see, feel, and hear certain elements. However, for the elements of an observation not able to be assessed accurately by the human eye, there are handheld instruments available to an observer to gather the data of those elements. The most common instrument used to obtain this data is the Kestrel.



Exercise / Practice: Types of Observations

Operating the Kestrel

Kestrel sensors measure multiple atmospheric parameters. When calibrated correctly, it provides accurate, real-time data for wind data, temperature and dew point, pressure values, elevation, and compass direction.





Wind Data: Perform the following steps to obtain wind direction and speed

- Face into the wind
- Access the compass feature and hold Kestrel at eye level
- Read display in degrees and round direction to nearest 10°
- Access the speed feature
- Monitor display to obtain 2-minute average speed (10-minute average for gusts)
- Round to the nearest whole knot

Temperature: Use the temperature feature to obtain the temperature and dew point to the nearest whole degree Celsius. Ensure the recorded dew point is equal to or less than the temperature value.

Pressure: Select the appropriate pressure feature (Pressure = Station Pressure / Baro = Altimeter Setting). Round the station pressure down to the nearest .005 inch or the altimeter setting to the nearest .01 inch.

Lastly, you should determine all applicable remarks and additive data to complete the observation. Included all phenomena related to outside weather elements.

Exercise / Practice: Operating the Kestrel **Mandatory Observation Elements**

METAR KBIX 241155Z 24010G20KT 2 1/2SM R23/6000FT +TSRA
BR SCT020CB BKN070 OVC200 01/M01 A2992

Station Identifier

The International Civil Aviation Organization (ICAO) designates all station identifiers using four alphanumeric characters. The first letter represents the airfield's aeronautical fixed service routing area. The remaining three characters specify the observation site. For example, KBIX is the station identifier for Keesler AFB, Mississippi.

Date/Time of Observation

This is the date and time of the observation in Coordinated Universal Time (UTC), which is the current time at the prime meridian, or the zero-degree longitude line. It is the global standard for all observation sites, and it is referred to as "Zulu" (Z). The day of the month is the first two digits, and the remaining four numbers is the observation in hours (second two numbers) and minutes (last two numbers). The last character (Z) represents the standard, UTC time.

Exercise / Practice: ICAO & Date/Time

Wind Data

You learned in the previous unit how wind data is measured and reported. Within an observation, the wind data is gathered by using a sensors 2-minute average for speed. The only additional information needed for how to encode wind data in an observation is wind gusts. Wind gusts, if reported, are encoded the maximum wind speed (gust) following the sustained wind speed and the letter "G". When encoding wind speed and gusts, you must include their units of measure. An entry of 03025G50KT would indicate wind from 30 degrees at 25KT gusting to 50KT.

Exercise / Practice: Wind Data

Prevailing Visibility

Visibility is a measure of the opacity of the atmosphere and is expressed in terms of the horizontal distance at which a person can see and identify specified objects. Visibility values are reported in statute miles for CONUS and meters for OCONUS. A visibility binder is a collection of labeled photographs, lists or other means of identifying objects with known distances to be used as aids when manually determining visibility ranges from the primary observation point.

Runway Visual Range (RVR)

RVR is an instrumentally derived value that represents the horizontal distance that a pilot can see down the runway. The maximum distance in the direction of takeoff or landing at which the runway, or specified lights or markers delineating it, can be seen from a position above a specified point on its center. For all locations reporting RVR, the unit of measure will correspond to the units of measure used to report visibility.

Exercise / Practice: Visibility Data

Present Weather

Present weather refers to any weather element occurring within 5 SM of the point of observation. Reports may be instrument derived, physically observed or determined using a combination of the two. Present weather is not in an observation when visibility is unrestricted. When visibility

is restricted, present weather may include one weather group or multiple weather groups found in your branch's observation manual. The present weather group has two sections: qualifier, which is phenomena characteristics, and weather phenomena separated into precipitation, obscurations to vision, and other phenomena.

Exercise / Practice: Present Weather Data

Sky Condition

Sky condition is the amount of clouds and obscuring phenomena covering the celestial dome. The celestial dome is the amount of sky not including any obstructions such as trees, mountains or human-made structures.

Layer Amount: The celestial dome is split into eights or "octas". The standard format for sky condition includes a contraction for the layer amount, layer height, and significant cloud type. Each layer requires its own group so the observation may include one or more than one group. Sky condition is encoded in ascending order up to the first overcast layer using the summation principle. Layer amount refers to the number of octas a certain layer of clouds cover. Layer amounts are clear (CLR or SKC / no clouds present), few (FEW 1/8 – 2/8 coverage), scattered (SCT 4/8 – 5/8 coverage), broken (BKN 5/8 – less than total coverage), and overcast (OVC / total sky coverage).

Decoding Sky Condition: When decoding sky condition from a METAR or SPECI report, the layer height refers to the altitude in feet above ground level (AGL) where the base of a cloud layer begins. It is encoded in hundreds of feet using a three-digit format:

- 001 = 100 feet
- 010 = 1,000 feet
- 100 = 10,000 feet

This format allows observers and forecasters to quickly interpret cloud base heights from encoded data.

When encoding sky condition into a METAR or SPECI, cloud base heights must be rounded down to the nearest reportable increment based on altitude range. This ensures consistency and standardization across all observations.

Altitude Range (AGL) Reportable Increment Example

- 0–6,000 ft Nearest 100 ft 2,475 ft → 2,400 ft → 024
- 6,001–10,000 ft Nearest 500 ft 6,750 ft → 6,500 ft → 065
- Above 10,000 ft Nearest 1,000 ft 11,800 ft → 11,000 ft → 110

Always round down, never up—even if the measured value is close to the next increment.

Layer Height: Layer height indicates at what height in feet the base of each cloud layer starts. Layer heights are in hundreds of feet at ground level (AGL) in a three-digit format. For example, a code of 001 equals 100 feet, a code of 010 is 1,000 feet, and 100 equals 10,000 feet.

Significant Cloud Type: Cumulonimbus (CB) or towering cumulus (TCU) clouds are the only significant cloud types so it may or may not be included in the observation. Some examples of proper decoding are:

- FEW080 – Few coverage with bases at 8,000 feet
- SCT025CB – Scattered coverage with bases at 2,500 feet with a cumulonimbus cloud observed.
- FEW007 BKN025 OVC110 – Few coverage with bases at 700 feet, broken coverage beginning at 2,500 feet, and overcast at 11,000 feet

There are two coverages for a clear sky. The difference between them is an automated observing station reports CLR when there are no clouds between the sensor and 12,000 feet while a manual observing station reports SKC when there are no clouds in the celestial dome.

Exercise / Practice: Sky Condition

Ambient Air and Dew-Point Temperatures

Ambient air and dew-point temperatures are two digits in whole degree Celsius (°C) separated by a solidus (/). When the letter "M" precedes the number, the temperature is below freezing. For example: 04/M02 – Temperature is 4°C and dew point is negative 2°C.

Exercise / Practice: Temp / DP Data

Pressure

METAR observations in CONUS report barometric pressure using altimeter setting where "A" is the altimeter-setting indicator and is the pressure to the nearest hundredths of an inch of mercury without the decimal (e.g., A2992 = 29.92 inches of mercury). Alternatively, OCONUS observations report pressure in millibars and will be encoded in the observation beginning with a "Q" and pressure is reported to the nearest whole millibar without the decimal. (e.g., Q1013 = 1,013 millibars).

Station Pressure: Station pressure is the atmospheric pressure at a given elevation. It is the standard pressure for all other pressure values.

Altimeter Setting (ALSTG): Altimeter setting is a conversion of station pressure to MSL using inches of mercury. Aviators (pilots) use it during takeoff and landing.

Sea-level Pressure (SLP): Sea-level pressure is a conversion of station pressure to MSL using millibars. Meteorologists use it to depict high- and low-pressure systems on a surface analysis chart.

Exercise / Practice: Pressure Data

Remarks

You learned previously that the remarks section of an observation is used to elaborate on any parameters the observer notes as important information or is directed to include based on the

requirements laid out in the remarks section of the observing manual. The remarks attachment of the observing manuals are numbered and listed in the order of importance and the order in which they will be recorded in the observation. With each remark, there is instructions and examples on how to encode the given element. There is also guidance on whether the remark is mandatory, optional, or not required depending on the type and situation of the observed condition.

Exercise / Practice: Remarks

Taking a Surface Weather Observation



Timing

The accuracy of the time ascribed to weather observations is of the utmost importance. You will encode all METAR observations between 55 and 59 minutes past the hour. All SPECI observations will be encoded with the time the weather event was observed and recorded. For example, “lightning within 5sm was confirmed via sensor at 32 minutes past 1400 on the 28th day of the month. Your SPECI time would read. SPECI 281432Z

Materials & Resources

Before taking an observation, you must ensure you have all the required materials and resources needed. Some items needed might include:

- A pencil to write with and paper to write on (official form or notebook paper)
- A kestrel or other measuring device if taking a manual observation
- Sunglasses / eye protection
- Reference Manuals
- Ruler or other measuring devices if needing to record hail size.

If you take an augmented observation, you will also need to ensure you have access to the airfield sensor data to acquire the necessary data.

Location

All official observations are taken at the same point at every airfield. Operationally, this official observation point has certain specifications that you will learn about once you leave the training environment. For the purposes of this course, the official observation point is anywhere on the observation deck of the Weather Training Complex.

Recording & Submitting Observations

Most observations are recorded and submitted using an online source that you will get familiar with later in the course. However, depending on resources available and potential communications disruption, each military service has their own official forms to record their observations. During this portion of the course, you will record your observations manually on the paper form.

- **Air Force – Form 3803 / Navy/Marines – Form 3140**

Regardless of which form you use, all observation entries must be legible and written in pencil. Another requirement is to use capital, block letters and numbers. Air Force students should encode SA for METAR observation and SP for SPECI observations. Navy and Marine Corps students encode M for METAR observations and an S for SPECI observations.

Exercise / Practice: Taking an Observation

Practice: Observations

PC: Decoding / Encoding / Taking Observations